# Research Brief | Wells With Well Log Information

*New Mexico Office of the State Engineer (OSE) • San Juan basin / San Juan sub-basin*

### Dataset and geographic relevance

The supplied CSV contains 1,379 well-log records, including 1,313 with borehole depth and 1,016 with depth where water was first encountered. County codes include San Juan (793), Rio Arriba (317), McKinley (260), and Sandoval (6); three records have unknown/other county codes. The file includes well identifiers, PLSS descriptions, coordinates, and drilling dates.

Relevance to Cuba: INDIRECT / NOT YET VERIFIED. Only six records are coded to Sandoval County, and the file does not identify Cuba's municipal wells or their producing geological aquifer. The OSE administrative basin filter should not be equated with the San Juan Structural Basin.

### Research-question support

| Question | Support | Use and limitation |
|---|---|---|
| **Q2 — Well characteristics** | Strongest (partial) | Compare borehole depths and initial water encounters; aquifer units not identified. |
| **Q4 — Aquifer comparison** | Potential | Individual detailed logs may help identify formations; CSV alone cannot compare Ojo Alamo vs. San Jose. |
| **Q8 — Data coverage** | Useful | Map well-log locations and missing fields; not a map of active groundwater monitoring. |
| **Q9 — Stress dashboard** | Supporting context | Show well-depth distributions and spatial coverage; not a depletion or stress metric. |
| **Q1 / Q3 — Trends / depletion** | Not supported | No repeated groundwater levels, withdrawal volumes, or recharge estimates. |

### Recommended plots for the project report

1. Map well locations by county (check coordinate reference system before plotting). 2. Boxplots or histograms of borehole depth by county. 3. Scatter plot of "Depth Water First Encountered" versus "Borehole Depth," excluding missing values. 4. Counts of available well logs by drilling year and county.

### Interpretation cautions and next step

"Depth Water First Encountered" is a drilling observation, not a present-day water-table reading. Well-log counts do not indicate monitoring density or pumping activity. To assess Cuba specifically, verify coordinates of the six Sandoval records, identify municipal well IDs, and request complete logs and time-series groundwater levels from OSE or the Village of Cuba.

*Source: OSE New Mexico Water Rights Reporting System, "Wells With Well Log Information" (user-provided CSV; San Juan basin and sub-basin filters).*